DICTIONARY-ASSISTED HYBRID RETRIEVAL-AUGMENTED GENERATION ASSISTANT FOR CODE-SWITCHED LOCAL GOVERNMENT SERVICES INQUIRIES IN DAVAO CITY

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BACKGROUND

1. Davao City's eBOSS (Electronic Business One-Stop Shop) processes thousands of citizen business permit queries annually.
2. Citizens in Davao City communicate in code-switched language.
3. Existing chatbot and FAQ systems fail to bridge the lexical gap between colloquial dialect queries and the formal statutory language used in Citizens' Charter documents.
4. There is no existing NLP-driven multilingual system purpose-built for Davao City local government service delivery.

LGS THESIS PROGRESS

THESIS STATEMENT

This study investigates whether dictionary-assisted multilingual query normalization combined with hybrid RAG retrieval improves access to relevant Davao City Citizens' Charter information compared with raw BM25 retrieval. It also examines whether the retrieved evidence supports relevant, faithful, complete, and language-consistent answers, while distinguishing the contribution of dictionary injection from the broader normalization and retrieval pipeline.

LGS THESIS PROGRESS

PROBLEM STATEMENT

1. How can a Dictionary-Assisted Hybrid RAG approach be implemented to capture both charter identifiers and conversational intent?

2. How does the proposed Dictionary-Assisted Hybrid RAG approach perform compared with a retrieval baseline in processing code-switched local government service inquiries?

3. To what extent are the responses generated by the proposed system relevant and factually faithful to the official Davao City Citizens' Charter, as measured through automated RAGAS metrics and human expert evaluation?

RESEARCH OBJECTIVES

1. To implement the Dictionary-Assisted Hybrid RAG approach to capture both charter identifiers and conversational intent.

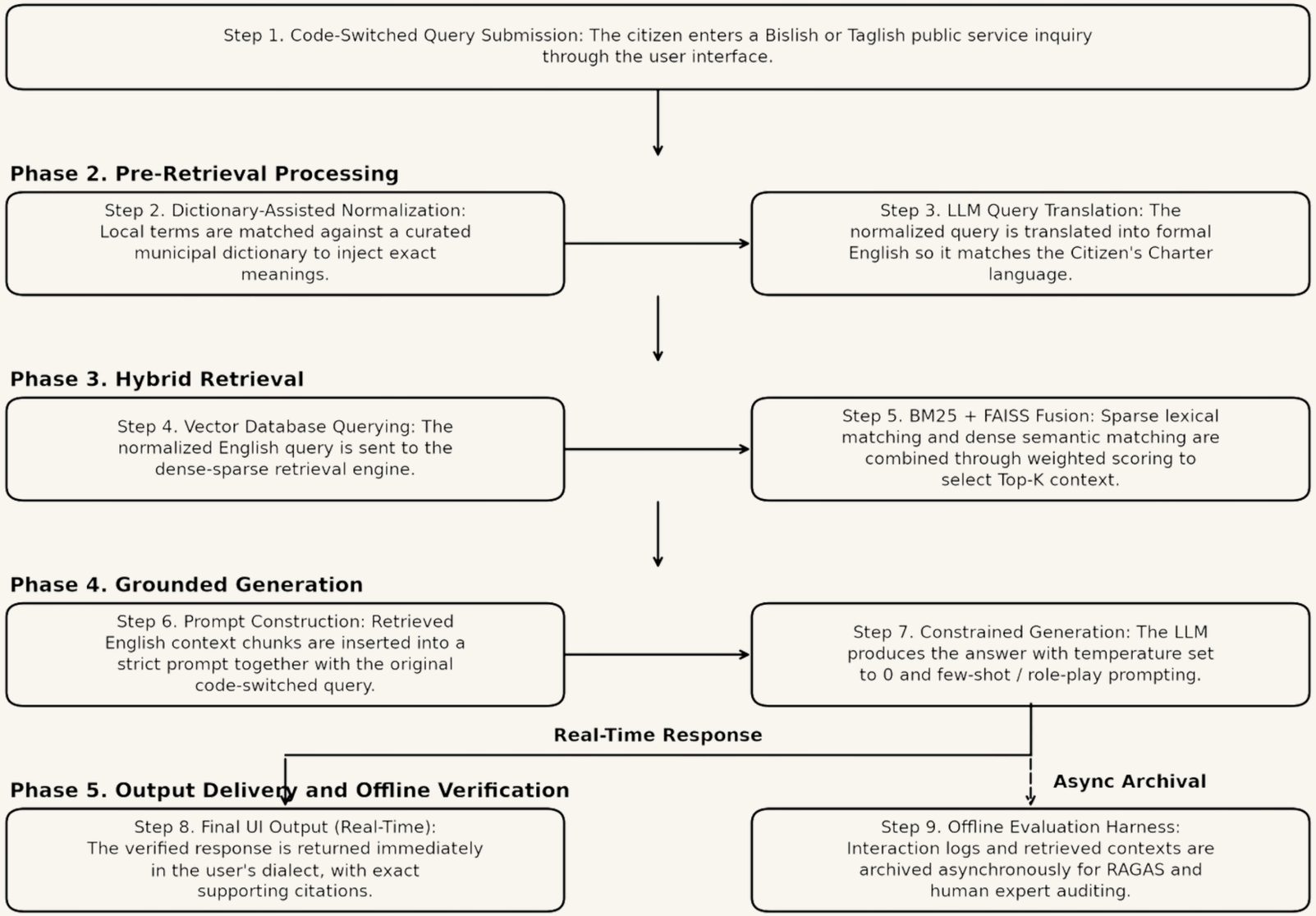
2. To compare the performance of the proposed Dictionary-Assisted Hybrid RAG approach with retrieval baselines in processing code-switched local government service inquiries.

3. To evaluate the answer relevance and factual faithfulness of the proposed system using RAGAS metrics and human expert assessment based on official Davao City government documents.

RESEARCH DESIGN OVERVIEW

1. Corpus extraction, dictionary curation, dual indexing, local LLM server deployment
2. 100-scenario benchmark authoring, development grid search, held-out test evaluation
3. Human expert auditing, Friedman/Wilcoxon statistical testing, final reporting.

CONCEPTUAL FRAMEWORK



RESEARCH DESIGN OVERVIEW

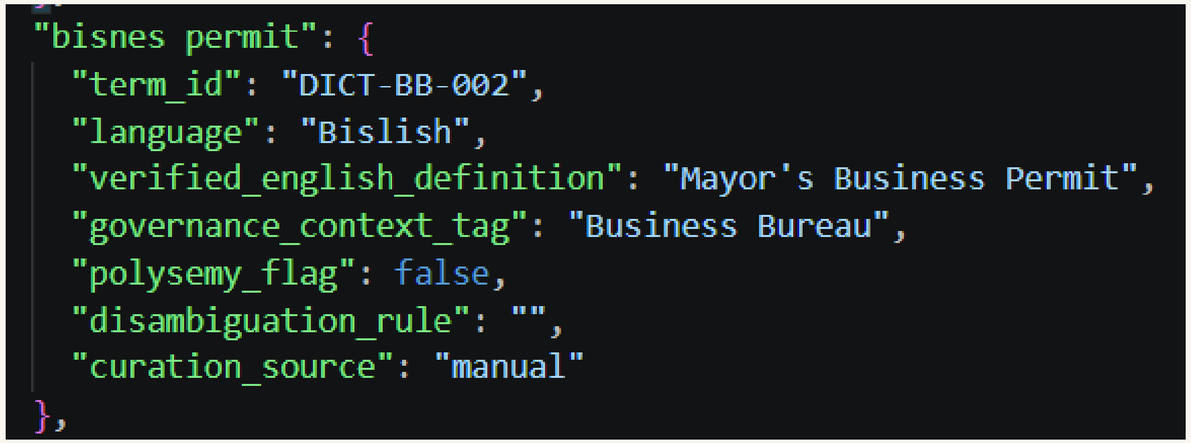
Construct the Localized Dictionary

- Layer 1: Manual Curation: 50 high-priority terms focused on ambiguous vocabulary and dialect false friends.
- Layer 2: Scraped and validated supplement from Davao City charter text and web corpus, subject to three-step validation
  (domain filter → disambiguation review → definition verification)

- Structured Schema
  - **local_term**: colloquial word/phrase
  - **verified_english_definition:** official statutory counterpart
  - **governance_context_tag:** target municipal office
  - **polysemy_flag:** true / false

RESEARCH DESIGN OVERVIEW

Construct the Localized Dictionary



```
"bisnes permit": {
  "term_id": "DICT-BB-002",
  "language": "Bislish",
  "verified_english_definition": "Mayor's Business Permit",
  "governance_context_tag": "Business Bureau",
  "polysemy_flag": false,
  "disambiguation_rule": "",
  "curation_source": "manual"
},
```

The final dictionary is saved as a structured JSON file. Each entry lists the local term, its identified source language, its verified English definition, its source of curation, either manual or from a specific scrape

RESEARCH DESIGN OVERVIEW

Corpus Description, Sampling Gaps, and Preprocessing Pipeline

- Corpus Sources: 5 Official Citizens' Charter documents covering 5 eBOSS pillars:
  - Business Bureau (250 chunks), City Treasurer's Office (220 chunks), OCBO (180 chunks), CPDO (190 chunks), BFP (191 chunks) = 1,031 Chunks Total.
  - Embedding model architecture constraint: paraphrase-multilingual-MiniLM-L12-v2 has a strict 128-token context window.
  - Selected chunk size: 120 tokens with 20-token overlap using HuggingFace tokenizer splitting (prevents silent truncation during vectorization).
- Three Inherent Source Biases in Raw Government Data:
  - **Uneven Service Coverage**: High-volume transactions (e.g., Mayor's permit renewals) dominate charter pages; rare procedural exceptions possess minimal textual evidence.
  - **Formal Institutional Language**: Heavy administrative phrasing overrides colloquial speech.
  - **Time & Versioning Bias**: Outdated charter schedules contain superseded fees/offices.



RESEARCH DESIGN OVERVIEW

Corpus Description, Sampling Gaps, and Preprocessing Pipeline

Requirement	How Handled in RAG Methodology
Data Imbalance & Sampling Gaps	**Stratified Benchmark Quota:** In RAG retrieval, synthetic oversampling (SMOTE) is strictly avoided because it corrupts factual text. Instead, bias is controlled at evaluation via an equal quota of **20 scenario groups per pillar** (100 total).
Digitization & Table Artifacts	**Tabular Integrity Cross-Check:** Procedural fee tables verified manually against original page images. LLMs are strictly forbidden from hallucinating missing text.
Versioning Bias	**Superseded Isolation:** Legacy schedules isolated to change logs; only current charters are indexed.
Overfitting Controls	**Inference-Only Architecture:** Pretrained transformer checkpoints operate in pure inference mode (no backpropagation/dropout training). Overfitting is controlled via a **strict pipeline freeze point** (pipeline_manifest.json) and double-blind test quarantine.



RESEARCH DESIGN OVERVIEW

Pre-Retrieval Query Normalization Pipeline

Dictionary-Assisted Lexical Injection

- Scans incoming query using word-boundary regex (\b) against municipal_dictionary_v1.json.
- Injects verified definitions directly into the translation prompt as grounding context, preventing probabilistic hallucination.

Constrained LLM Query Translation

- Executes few-shot translation via local Mistral 7B (eboss-mistral).
- Enforces a hard 150-token generation ceiling (num_predict=150) to physically eliminate overgeneration and conversational filler

Ambiguity Fallback Mechanism

- Intercepts terms flagged with polysemy_flag: true (e.g., terms with conflicting cross-departmental meanings).
- Halts retrieval and prompts citizen for clarification ("Did you mean A or B?"), preventing corrupted query vectors from polluting the index.

RESEARCH DESIGN OVERVIEW

Hybrid Retrieval Index

- **Sparse Component (BM25):** Inverted index over 1,031 tokenized chunks (data/bm25_index.pkl) for exact statutory acronyms, section numbers, and numerical fees.
- **Dense Component (FAISS):** 384-dimensional inner-product index (data/faiss_index.bin) using paraphrase-multilingual-MiniLM-L12-v2.

Min-Max Score Normalization:

$$\bar{S}(d) = \frac{S(d) - S_{\min}}{S_{\max} - S_{\min} + \epsilon}$$

Linear Weighted Fusion

$$\text{Score}_{\text{hybrid}}(d) = \alpha \cdot \bar{S}_{\text{BM25}}(d) + \beta \cdot \bar{S}_{\text{FAISS}}(d) + \gamma \cdot \bar{S}_{\text{Dict}}(d)$$

$$\text{subject to } \alpha + \beta + \gamma = 1.0$$



RESEARCH DESIGN OVERVIEW

Develop the LLM Generation Module

- **Model Hosting:** Locally served via Ollama (eboss-mistral) on dedicated GPU VRAM.
- **Deterministic Inference Constraints:**
  - Temperature=0.0 and Top_P=0.0Top_P=0.0 (Strict greedy decoding; guarantees identical inputs yield identical outputs).
  - Context Budget: K=3 retrieved chunks injected into the prompt.
- **Anti-Hallucination Prompt Architecture:**
  - Model is instructed: "Answer solely based on the provided context passages. If the required fee or step is not in the context, explicitly state that it is not specified in the active Citizens' Charter."
  - Requires citation of the exact issuing office and procedural step in every response.

RESEARCH DESIGN OVERVIEW

Web Application Architecture

- **Backend Architecture:** Built with FastAPI (uvicorn), hosting REST endpoints for query normalization, hybrid search, and generation.
  - **In-memory startup:** BM25 and FAISS indices preloaded at boot (one-time latency cost).
  - **Asynchronous logging:** Background recording of interactions for auditing.

- **Frontend Architecture:** Lightweight responsive chat interface:
  - Supports input in Bislish, Taglish, and English.
  - Interactive Disambiguation Modal: Renders the clarifying question when Fallback is triggered.
  - Displays retrieved source references alongside generated answers.

RESEARCH DESIGN OVERVIEW

Automated Evaluation Setup & Benchmark Design

- **100-Group Multilingual Benchmark (data/benchmark_scenarios_*.json):**
  - 100 scenario groups × 3 languages = 300 total queries (English, Taglish, Bislish).
  - Equal quota of 20 groups across all 5 eBOSS pillars.

- **Rigorous Partitioning:**
  - Development Set (30 Groups / 90 Queries): Used exclusively for prompt engineering and fusion weight grid search.
  - Held-Out Test Set (70 Groups / 210 Queries): Quarantined until final single-pass evaluation.

- **Three Comparative Experimental Configurations:**
  - Config A (BM25 Baseline): Raw unnormalized query → BM25 sparse retrieval only.
  - Config B (Proposed System): Normalization + Dictionary Injection + Hybrid Retrieval - Tuned via Validation Grid
  - Config C (Ablation — No Injection): Normalization WITHOUT Dictionary Injection + Hybrid Retrieval.

RESEARCH DESIGN OVERVIEW

Controls and Pipeline Safeguards

1. **Inapplicability of Traditional Supervised Training Controls**

2. **Five Rigorous Safeguards Against Pipeline Overfitting**
   - En Bloc Grouped Scenario Partitioning
   - Double-Blind Test Quarantine
   - Pipeline Freeze Point
   - Deterministic Inference
   - Unfiltered Performance Reporting



RESEARCH DESIGN OVERVIEW

Human Expert Auditing Framework

- **At least two (2) independent domain auditors from the LGU**
- **System identities masked; presentation order randomized to eliminate evaluation bias.**
- **Three-Component Evidence Record:**
  - Component 1: Ground Truth Reference (procedural steps, fees, timelines, page citations).
  - Component 2: Generated System Output (answer, normalized query, retrieved chunks).
  - Component 3: Human Audit Evaluation (independent score sheets, qualitative notes, error flags).
- **Audit Sample Allocation:**
  - 20 test scenario groups (4 per pillar × 3 languages = 60 queries).
  - Both Config A and Config B generate answers = 120 total system outputs.
  - 3 auditors ×× 120 responses = 360 completed audit evaluations.
- **4-Point Ordinal Scoring Rubric**

EVALUATION METRICS

Retrieval Recall & Rank Positioning

- **Hit@K →** Measures presence of ground-truth evidence

  Evaluated across cutoffs: $K \in \{1, 3, 5\}$

$$\text{Hit@}K = \mathbb{I}\left[\text{Top-}K \cap \text{GT} \neq \emptyset\right]$$

- **Mean Reciprocal Rank (MRR) →** Measures presence of ground-truth evidence

  Penalizes lower positions: $\frac{1}{\text{rank}_q}$

$$\text{MRR} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{\text{rank}_q}$$

EVALUATION METRICS

Context Precision (RAGAS Metric)

- Measures signal-to-noise density in prompt context
- Rewards ranking relevant chunks ahead of noise

$$\text{Precision@}k = \frac{\text{Relevant Chunks in Top } k}{k}$$

- Weighted by relevance indicator: $\text{rel}(k) \in \{0, 1\}$
- Mitigates LLM **"Lost-in-the-Middle"** degradation
- Selected as primary objective function: J(θ)

EVALUATION METRICS

Simplex Grid Search

- Discrete grid search across 17 weight candidates over K∈{3,5} (34 conditions) on 90 dev queries
- **Optimal Frozen Weights:**
  - **Sparse BM25: α = 0.75**
  - **Dense FAISS: β = 0.25**
  - **Dictionary Boost: y = 0.0, Context Budget: K = 3**
- **Dev Peak Scores:**

$$\text{Macro Context Precision } (J) = 0.8685 \ (\text{Hit@3} = 70.00\%, \ \text{MRR@3} = 0.6167)$$

STATISTICAL MODEL

- **Omnibus Test — Friedman Test (Non-Parametric Repeated Measures):**

$$H_{\text{Friedman}} = \frac{12}{nk(k+1)} \sum_{j=1}^{k} R_j^2 - 3n(k+1)$$

- (where n=210 queries, k=3configurations, Rj = column rank sums)
- Tests H0: There is no significant difference in retrieval distributions between Configurations A, B, and C.

- **Post-Hoc Pairwise Testing — Wilcoxon Signed-Rank Test:**
  - Applied to paired comparisons: A vs. B, A vs. C, and B vs. C.
  - Controlled using Bonferroni correction $\alpha = 0.05/3 = 0.0167$)

INITIAL RESULTS

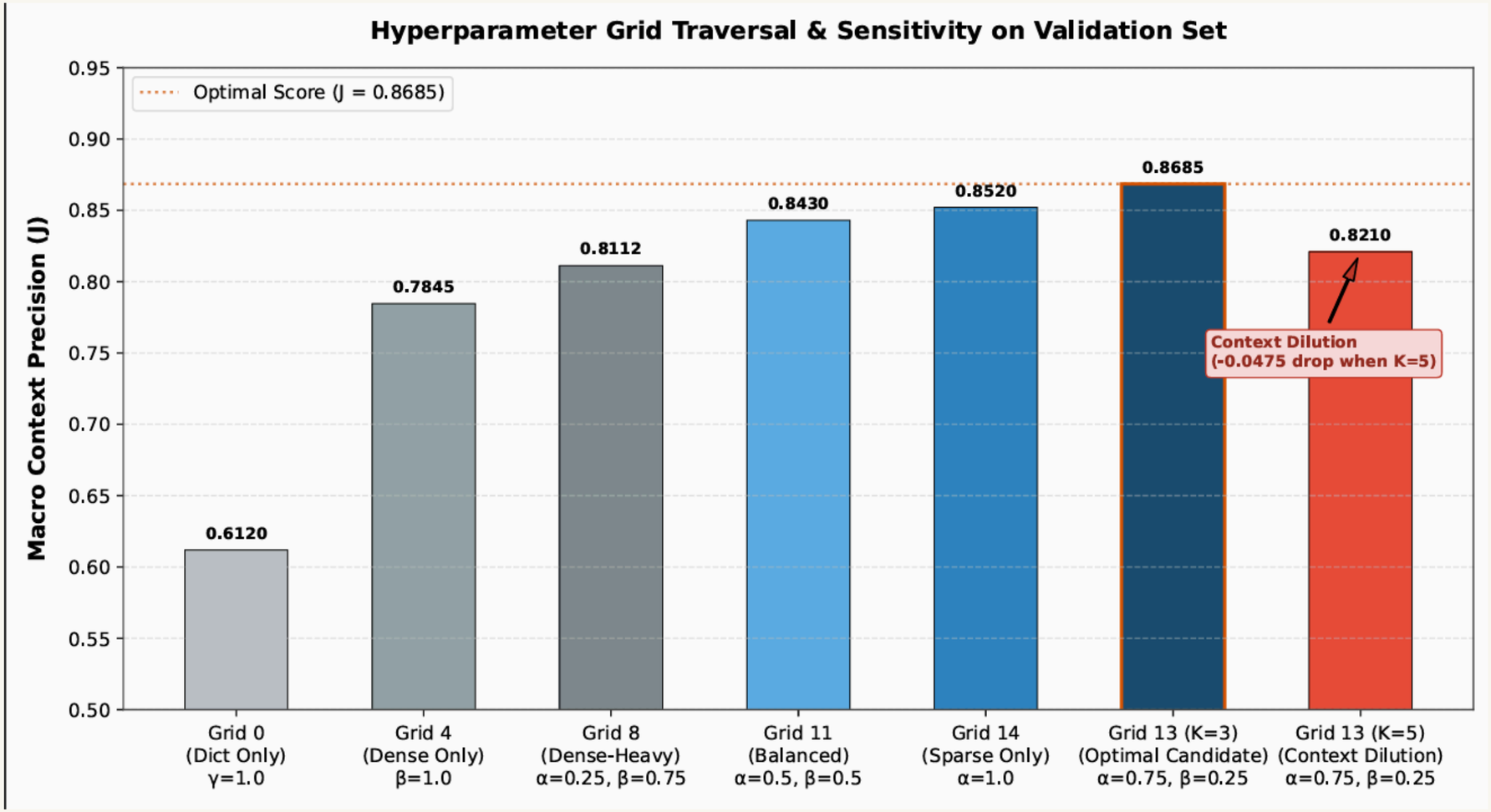


Figure 1: Hyperparameter Grid Traversal & Sensitivity on Validation Set

INITIAL RESULTS

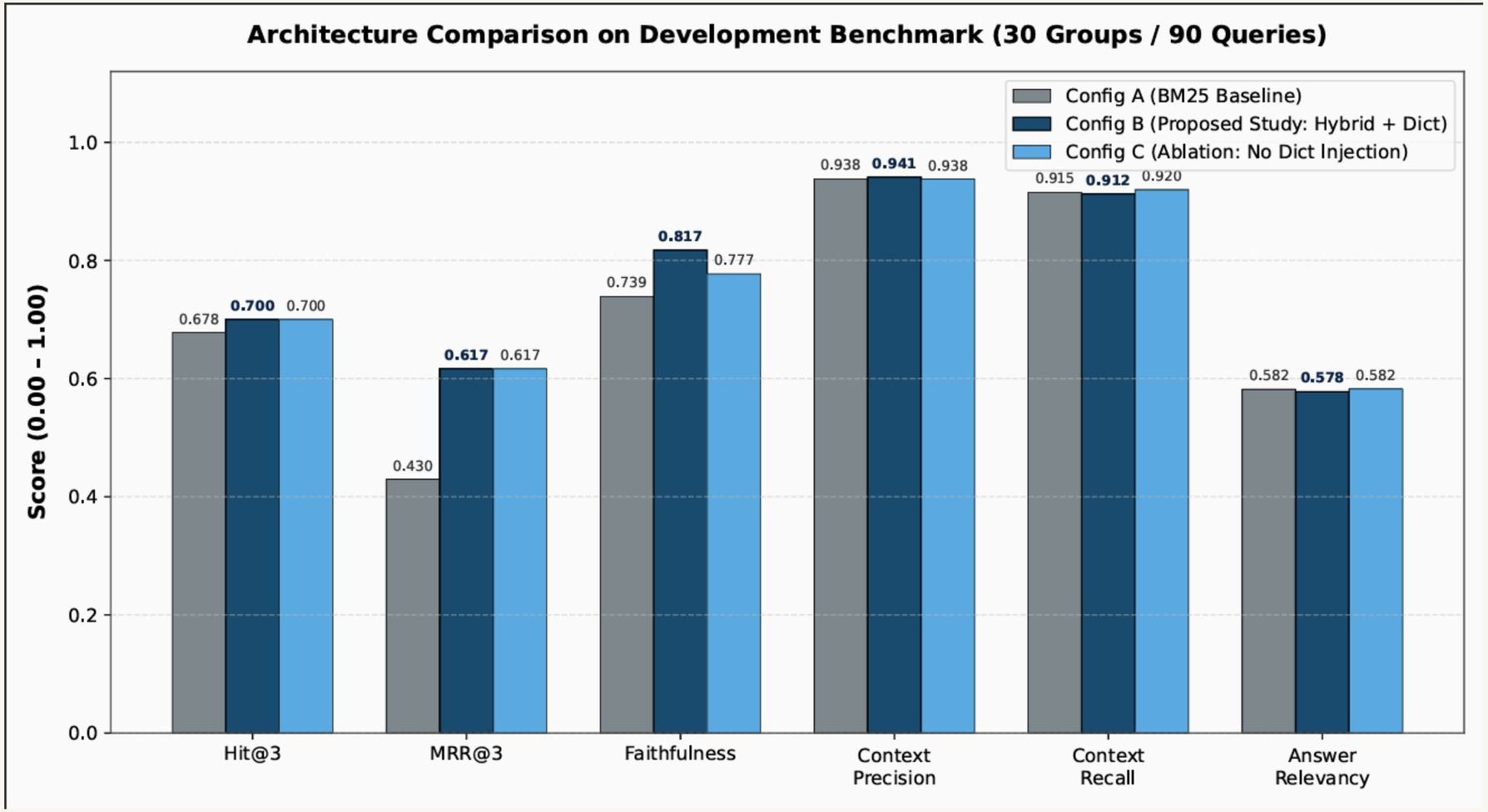


Figure 2: Architecture Comparison on Development Benchmark (30 Groups / 90 Queries)

IMPROVEMENT PRIORITIES

First: repair chunk IDs and section boundaries.

Improve retrieval using development data.

Enforce clarification before answering.

Preserve the user’s language and source conditions.

Return correct office metadata and supported citations.

Complete grouped inference and human evaluation.

